



Quickstart

The Home Connect API

This API provides access to home appliances connected to [Home Connect](#). Developers can use the Home Connect API to monitor and control these appliances. For instance, you could turn on your coffee machine and make your favorite cup of coffee. The following Quickstart guide will help you get started with exactly that.

Simulator vs. Home Connect API

"But how can I test my application?" you're asking. Of course you can use your very own Home Connect home appliance, with the actual Home Connect API, if you own one. In case you don't, we also provide a virtual environment with simulated appliances. This allows you to jump-start your development and test your applications. Please note that some of the actions you are able to perform on the simulators are not possible through the real API. One example of this is opening/closing doors. [You can find the simulators here.](#)

Using the simulator

Simulator

To use the simulator, log into your account and navigate to [Simulators](#). At the [Applications](#) page you will find an existing application called API Web Client. As you progress through the guide use the Client ID of this application and a scope of CoffeeMaker.

Host

<https://simulator.home-connect.com>

Using the real API



Host

<https://api.home-connect.com>

<https://api.home-connect.cn> (Greater China)

Authorization

For your application to interact with home appliances it needs to get authorized by the API. We will guide you through the steps required to complete the OAuth 2.0 Authorization Code Grant Flow process. Successful authorization will result in the API responding with a JSON Web Token.

Step 1: Get the Client ID

To get started, visit the [Home Connect Developer Portal](#) and register an account. After registration please access the [application overview page](#). Here you can find all your applications, including a predefined one called API Web Client.



Client ID

A54FDB23FFCC3B628F6DD092FF598C781524XD1
472A2663B6A94166X551D9C1D

API client - Simulated home appliances [↗](#)

API client - Physical home appliances [↗](#)

Access: Private

Publication Requests: 0

OAuth Flow: Authorization Code Grant Flow

Client ID

D368X79E8D91A13CE874F7DE226EC011C02ECA3
36D766A61B1F54B194AFXF62D

Details

Edit

Simulator

Use the Client ID of the API Web Client application and insert it into the query string of the URL.

Real API

To start testing on a real appliance, you should [register an application](#) first. Please choose the Authorization Code Grant Flow and fill out all required fields. Once done, complete the query string in the URL with your application's details.

Step 2: Authorization URL and Response

Next, call the *authorize* endpoint. Easiest way to do this is to copy the URL into your browser.

Simulator

```
https://simulator.home-connect.com/security/oauth/authorize?  
response_type=code&client_id=  
{client_id}&scope=IdentifyAppliance%20CoffeeMaker&redirect_uri={redirect_uri}
```

Try it out

Real API



When prompted, log in with your Home Connect Account and grant access. The API will respond with a HTTP 302 redirect to the defined `redirect_uri`. A request parameter in this response will contain the authorization code.

Response

```
https://example.com/?code=eyJ4LWVudiI6IlBSRCIsImFsZyI6IlJTMjU2IiwieC1yZWciOiJFV'
```

Step 3: Retrieve Token

Call the token endpoint including the authorization code obtained in previous step:

Simulator

```
curl -X POST "https://simulator.home-connect.com/security/oauth/token" \
-H "Content-Type: application/x-www-form-urlencoded" \
--data-urlencode "client_id={client_id}" \
--data-urlencode "redirect_uri={redirect_uri}" \
--data-urlencode "grant_type=authorization_code" \
--data-urlencode "code={code}"
```

Run previous simulator steps first

Real API

```
curl -X POST "https://api.home-connect.com/security/oauth/token" \
-H "Content-Type: application/x-www-form-urlencoded" \
--data-urlencode "client_id={client_id}" \
--data-urlencode "client_secret={client_secret}" \
--data-urlencode "redirect_uri={redirect_uri}" \
--data-urlencode "grant_type=authorization_code" \
--data-urlencode "code={code}"
```

Try it out

Response



Conclusion

You should now have a valid access token which enables you to use the API!
For more detailed information about the authorization process visit the [dedicated page](#).

Monitoring

There are a few common use cases when developing an application with the Home Connect API. Displaying live status information of home appliances is one of them. Let's find out how to do that.

Step 1: Choose an Appliance to Use

To identify our home appliances, let us first request a list of all currently paired devices:

Simulator

```
curl -X GET "https://simulator.home-connect.com/api/homeappliances" \  
-H "Authorization: Bearer {access_token}"
```

Run previous simulator steps first

Real API

```
curl -X GET "https://api.home-connect.com/api/homeappliances" \  
-H "Authorization: Bearer {access_token}"
```



Response

[Expand source](#)

Step 2: Retrieve the Operation State

Simulator

```
curl -X GET "https://simulator.home-
connect.com/api/homeappliances/{hald}/status/BSH.Common.Status.Operation
State" \
-H "Authorization: Bearer {access_token}"
```

Run previous simulator steps first

Real API

```
curl -X GET "https://api.home-
connect.com/api/homeappliances/{hald}/status/BSH.Common.Status.Operation
State" \
-H "Authorization: Bearer {access_token}"
```

Try it out

As simple as that, you can retrieve the current operation state of your home appliance. You can read more about status information on the [states page](#).

Response

```
{
  "data": {
    "key": "BSH.Common.Status.OperationState",
    "value": "BSH.Common.EnumType.OperationState.Ready"
  }
}
```



Controlling

Monitoring your home appliances is pretty neat. What's even more neat though is being able to control it. So to conclude the Quickstart, we're going to find out how to toggle our home appliances on/off and start a program.

Step 1: Turn On

The power state setting of a home appliance is what tells you whether the appliance is on or off. In general, you will be able to write this setting to toggle it. Some appliances don't have that capability. For more detailed information on which appliances do, visit the [power state page](#).

A response of `204 - No Content` is a sign the operation was successful. In that case your coffee machine should turn on and start rinsing.

Simulator

```
curl -X PUT https://simulator.home-
connect.com/api/homeappliances/{hald}/settings/BSH.Common.Setting.PowerS
tate \
-H "Authorization: Bearer {access_token}" \
-H "Content-Type: application/vnd.bsh.sdk.v1+json" \
--data-raw '{
  "data": {
    "key": "BSH.Common.Setting.PowerState",
    "value": "BSH.Common.EnumType.PowerState.On"
  }
}'
```

Run previous simulator steps first

Real API

```
curl -X PUT https://api.home-
connect.com/api/homeappliances/{hald}/settings/BSH.Common.Setting.PowerS
tate \
-H "Authorization: Bearer {access_token}" \
```



```
}  
'
```

Try it out

Step 2: Start Program

And finally we will instruct the coffee machine to make 35ml of strong Espresso. You can find the coffee machine programs and options in detail [here](#).

Once again, a response of `204 - No Content` means it worked. Go to your coffee machine to find yourself a freshly brewed Espresso

Simulator

```
curl -X PUT https://simulator.home-  
connect.com/api/homeappliances/{hald}/programs/active \  
-H "Authorization: Bearer {access_token}" \  
-H "Content-Type: application/vnd.bsh.sdk.v1+json" \  
--data-raw '{  
  "data": {  
    "key": "ConsumerProducts.CoffeeMaker.Program.Beverage.Espresso",  
    "options": [  
      {  
        "key": "ConsumerProducts.CoffeeMaker.Option.BeanAmount",  
        "value": "ConsumerProducts.CoffeeMaker.EnumType.BeanAmount.Strong"  
      },  
      {  
        "key": "ConsumerProducts.CoffeeMaker.Option.FillQuantity",  
        "value": 35  
      }  
    ]  
  }  
'
```

Run previous simulator steps first

Real API



```
"key": "ConsumerProducts.CoffeeMaker.Program.Beverage.Espresso",
"options": [
  {
    "key": "ConsumerProducts.CoffeeMaker.Option.BeanAmount",
    "value": "ConsumerProducts.CoffeeMaker.EnumType.BeanAmount.Strong"
  },
  {
    "key": "ConsumerProducts.CoffeeMaker.Option.FillQuantity",
    "value": 35
  }
]
}
```

Try it out

Conclusion

Congrats on making it to the end of the Quickstart. You now know the basics of the Home Connect API and we cannot wait to see what new applications you're coming up with!

Next Chapter